Pavan B N

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LinkedIn | GitHub | TUF | LeetCode | Portfolio

WORK EXPERIENCE

IBM **Bengaluru, India**
Associate System Engineer (Application Developer) *Jun 2026 – Present*

- Build **2 Spring Boot microservices** and their **REST APIs** for an application modernization and migration program, delivering full-stack features across **Java, Spring Boot, and React**, and own their deployment via **Docker** and **Kubernetes**.
- Cut API response time from **200ms to 20ms** with **Redis** caching of frequently accessed data, and integrated **Apache Kafka** across **5 topics** to decouple services through asynchronous, event-driven processing.

Microsoft **Bengaluru, India**
Software Engineer Apprentice *Nov 2025 – Jun 2026*

- Optimized a Microsoft Fabric FMLV **REST API** that slowed as iterations grew: weighed caching against **batch initialization** with a Principal Engineer, shipped the more scalable option with tests, and cut lookups from **O(n) to O(1)**.
- Built the **functional test framework** for Multi-Schedule MLV Public APIs in **CI/CD**; used logs and **KQL** to root-cause a parallel-test **race condition** and fixed it with class-level batch cleanup instead of distributed locks, ending the flaky failures and **ICM** alerts; also fixed a **500**-instead-of-**404** bug before release.
- Remediated **70+ security vulnerabilities (CVEs)** in Apache Ambari (HDInsight) by tracing transitive dependencies and upgrading packages, with **zero functional regressions** in a legacy codebase.
- Delivered **GitHub Copilot** instruction files for **20+ repositories** in about 5 hours, using Copilot Agent drafts with full manual review.

Blink Health **Bengaluru, India**
Automation Engineer Intern *Jul 2024 – Jun 2025*

- Replaced browser-driven E2E validation with a reusable **HTTP client automation framework** across **100+ API endpoints**, raising automation reliability from **70% to 100%** and cutting execution time by **40%**.
- Proposed and proved out **Twilio's delivery-status API** for SMS tests, validating real customer delivery instead of database records; it became the team standard.

PROJECTS

Readora (Cloud-Native, AI-Powered Bookstore Platform) GitHub

- Architected an event-driven platform of **10 Spring Boot microservices** behind **Spring Cloud Gateway** with JWT authentication and Redis-backed rate limiting; coordinated **10 Kafka topics** using the transactional **outbox pattern** with idempotent consumers, circuit breakers, Redis caching, and **Flyway**-managed migrations.
- Built an **MCP server** exposing catalog, cart, and order data as tool-callable endpoints for a **RAG** reading assistant (**Azure OpenAI** + pgvector semantic search); shipped **3 React + TypeScript** applications using **Redux Toolkit** and **STOMP/SockJS** for real-time order and notification updates.
- Delivered a **GitOps** pipeline of **13 GitHub Actions** workflows that enforce a **90% JaCoCo coverage gate** (JUnit 5, **Testcontainers**), security-scan images, and push to **Azure Container Registry**, while **ArgoCD** auto-syncs an **AKS** cluster with **OpenTelemetry** distributed tracing to New Relic.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, SQL
Backend: Spring Boot, REST APIs, Microservices, Apache Kafka, Redis
Frontend: React.js, Redux, HTML5, CSS3
Databases: PostgreSQL, MySQL
Cloud & DevOps: Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD, KQL
AI: LLMs, RAG, Agentic AI, Model Context Protocol (MCP), Azure OpenAI, GitHub Copilot
Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Distributed Systems, System Design

EDUCATION

Reva University **Bengaluru, India**
B.Tech in Computer Science and Engineering (GPA: 9.16 / 10) *Oct 2021 – May 2025*

ACHIEVEMENTS & CERTIFICATIONS

- Solved **500+ DSA problems** across **LeetCode, Codeforces, GeeksforGeeks, and TUF** (Graphs, DP, Trees).
- Built Agentic AI and cloud-native solutions at **Microsoft Agent League**, **SIH**, **CodeFury**, and **Reva Hack**; completed Google's **Build with AI Bootcamp** (AI agents, agent workflows) and earned the **Google Cloud Build with Gemini** skill badge (Intermediate).
- Certified: **Claude Certified Developer (CCDV)**, **Microsoft Azure Fundamentals (AZ-900)**, and **Microsoft Azure AI Fundamentals (AI-900)**.